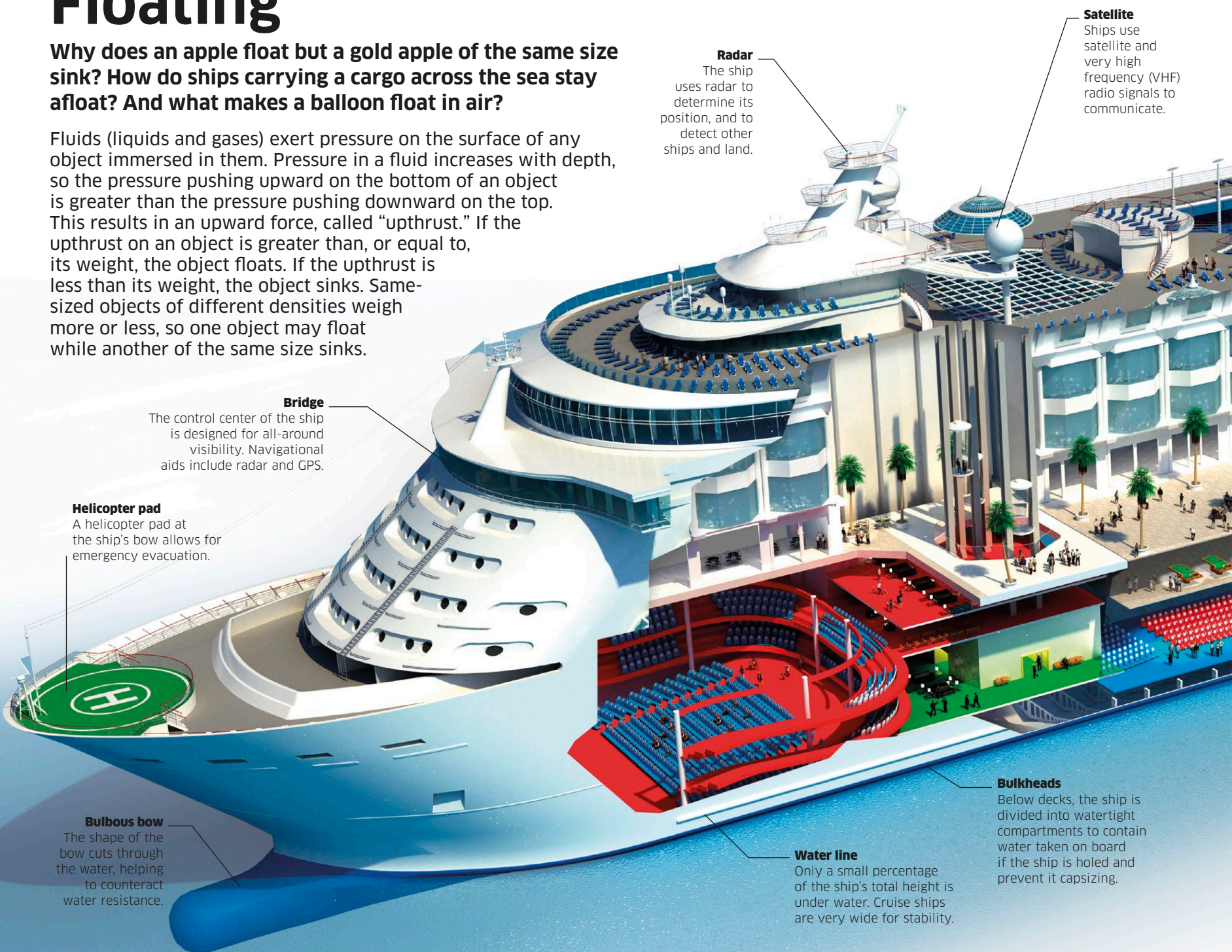


Floating

Why does an apple float but a gold apple of the same size sink? How do ships carrying a cargo across the sea stay afloat? And what makes a balloon float in air?

Fluids (liquids and gases) exert pressure on the surface of any object immersed in them. Pressure in a fluid increases with depth, so the pressure pushing upward on the bottom of an object is greater than the pressure pushing downward on the top. This results in an upward force, called “upthrust.” If the upthrust on an object is greater than, or equal to, its weight, the object floats. If the upthrust is less than its weight, the object sinks. Same-sized objects of different densities weigh more or less, so one object may float while another of the same size sinks.



Radar
The ship uses radar to determine its position, and to detect other ships and land.

Satellite
Ships use satellite and very high frequency (VHF) radio signals to communicate.

Bridge
The control center of the ship is designed for all-around visibility. Navigational aids include radar and GPS.

Helicopter pad
A helicopter pad at the ship's bow allows for emergency evacuation.

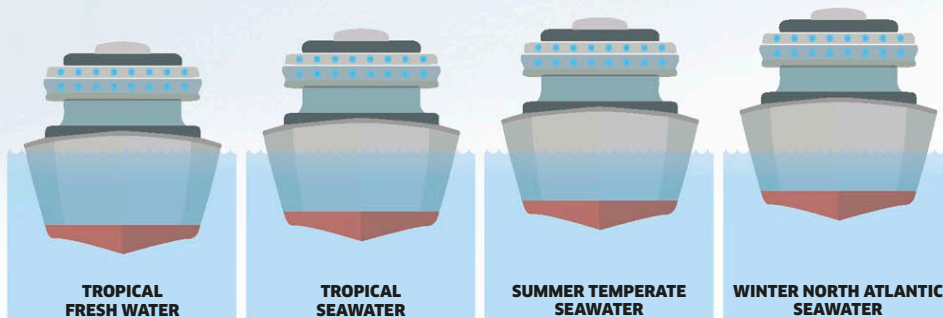
Bulbous bow
The shape of the bow cuts through the water, helping to counteract water resistance.

Water line
Only a small percentage of the ship's total height is under water. Cruise ships are very wide for stability.

Bulkheads
Below decks, the ship is divided into watertight compartments to contain water taken on board if the ship is holed and prevent it capsizing.

Water density

When ocean trade routes opened up around the globe, sailors were surprised to find their carefully loaded ships sank when they got near the equator. This was because the density of warm tropical waters was less than that of cool northern waters, and so provided less upthrust. When the ships entered freshwater ports, the water density was lower still, and ships were even more likely to sink.

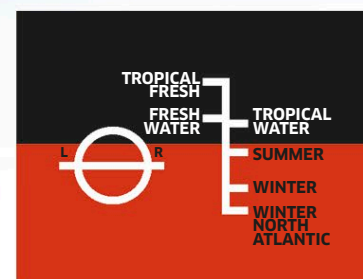


Warm water that is not salt has a low density, so a ship floats low in it.

Salt water has a higher density than fresh water, so a ship floats higher.

As salt water cools, its density decreases and a ship becomes more buoyant.

In the freezing cold waters of the North Atlantic, ships float high in the water.



The Plimsoll line

On a ship's hull, this mark shows the depth to which the ship may be immersed when loaded. This varies with a ship's size, type of cargo, time of year, and the water densities in port and at sea.